Petro Ladkin

Product Engineer. I take products from idea - or from stuck - to production. Robotics, backend, mobile; now with AI agents.

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SUMMARY

First program in 1995; paid to write software since 2006. I take a product at any stage - an idea, an MVP, a stalled project, an empty repo, a domain I have never seen - to production. Today I own intent, specs, acceptance and release, and direct AI agents to build and review what is needed.

SKILLS

Now	Product engineering / ROS 1+2 / C++ / Go / .NET (C#) / AI agents (specs, acceptance, agent review)
Also	Python / embedded (STM32) / iOS, macOS / security and VoIP / code generation / small teams
Earlier	Windows Mobile / Windows desktop (WinAPI, WTL) / Unity, AR

EXPERIENCE

Deus Robotics | Remote

Product Engineer — Apr 2026 - present

I own intent, specs, acceptance and release; AI agents build and review to that.

- Moved my own engineering to AI agents: in October 2026, 94% of my commits are agent co-authored. Built a multi-model code-review command - 6 specialist agents plus an arbiter - used in four repos.
- Rewrote the fleet operator web UI in React in 4 calendar days with a team of AI agents, replacing a 73.7k-line Flutter app; parity checked against 237 items of the old UI.
- Task dispatcher for a client delivery-robot project, sole author: first working version 9 days after the first commit, a full service with 1,097 tests in about 6.5 weeks (Go, design-first OpenAPI, PostgreSQL).
- Still maintainer and release owner of the fleet-management API (.NET): its unit-test suite reached 2,271 test methods.

Go / .NET (C#) / React / PostgreSQL / OpenAPI / AI agents

Senior Software Engineer — Oct 2021 - Apr 2026

Robot software in ROS 2, from first prototypes to the fleet backend.

- Learned ROS, which I had never seen, and had a layout-marking robot MVP driving on the map in about two months, navigating by a total station.
- Co-led the migration of the robot software from ROS 1 to ROS 2 without stopping the fleet: built the foundation (bridge proof of concept, multi-repo workspace, Docker, CI) and was the primary author of 10 of the 16 robot-side components.
- Built from an empty repo the adapter between ROS 2 and the company's own robots and chargers - a C++20 TCP server speaking their binary protocol - and took it to release 1.0 in ten months and 2.0 the next year; 358 test cases.
- Became maintainer and release owner of the fleet-management API (.NET microservices): migrated it to .NET 8 and grew its unit-test suite from 7 to 1,753 test methods.

ROS 1+2 / C++ / Python / C# / Go / CAN / CANopen / Modbus / ODrive / Docker / GitLab CI

Contracts and own product | Remote — 2020-2022

- **Staff-Eye GmbH, part-time:** refactored a client's production test scripts without access to their runner; an OMG IDL code generator to six targets, 1.0.0 in four months; ported a stalled STM32 firmware to F303.
- **CleverBooks:** rebuilt Space AR after its latest source code was lost; shipped Blocks AR to three stores.
- **AR agency:** Live History, an AR guide with lip-synced characters at 10 locations, and Kaup - each built alone from scratch to release.
- **RC Car in AR, own product:** my first Unity project, alone end to end: two months from the first commit to the App Store and Google Play.

Auxoft LLC | Chernivtsi, Ukraine / remote 2015-2020

Co-founder & CTO

KeyReel: a Bluetooth password manager - credentials stay encrypted on the phone and are sent to login pages when the phone is near the computer. **Pitched the startup in one call with a start budget I wrote myself; the condition was a proof of concept in 2 months - I built it, and the MVP got funded.**

- Took KeyReel from a 4-month MVP to a product on five platforms: iOS, macOS, Windows, browser extensions and Android.
- As CTO owned every technical decision - languages, frameworks, process; wrote 96% of the 68.5k-line Apple apps and 5,798 of the company's 8,917 commits.
- Built 2FA and U2F into KeyReel: an authenticator inside the password database, and the phone acting as a security key through a macOS driver.
- Set up the release process on my own git-flow CLI: two-week sprints, nightly to release builds - about 450 tagged builds.

Objective-C / Swift / C++ / Cocoa / UIKit / CoreBluetooth / IOKit / Thrift / JavaScript

Iosoftware | Chernivtsi, Ukraine / remote 2012-2014, part-time

Software Engineer

iOS apps for a small startup. **As lead developer, took BattleHorn, a castle-defense game, from a one-battlefield MVP to an App Store release in autumn 2013.**

- Came in for small graphics fixes in JukeMaster, an iOS DJ app, and found why scratching failed: the UI and audio threads blocked each other. Rewrote that part; the gig became a direct contract.

Objective-C / C++ / UIKit / AudioQueue / Cocos2d-x / RakNet

Freelance on oDesk | Chernivtsi, Ukraine / remote 2008-2012, part-time

Software Engineer

Windows Mobile apps and games, alongside the full-time job. **Took over PhoneyCall mid-development - a custom, undocumented UI framework over raw WinAPI, no access to its author - and shipped it in 137 hours.**

- Client rating 4.4 of 5 on both rated contracts - with almost no English at the time.

C++ / C# / WinAPI / WTL / .NET CF / TAPI / MAPI

SecurStar GmbH | Chernivtsi, Ukraine 2006-2015

Junior to Senior Software Engineer, System Architect, Team Lead

Encrypted calls for mobile phones. **When our main customers' network dropped the old protocol, built a new P2P-based product in 4 months; customers moved over.**

- As a junior, rewrote SecurePhone from C# to C++ and shipped it.
- PhoneCrypt: MVP, authentication and media servers (NAT traversal), Windows Mobile and iOS clients; the iOS core was still in production ~9 years after I left.
- Team Lead of 6 engineers (3 senior); 2 of my 3 juniors grew to middle.
- Release automation; moved the team from SourceSafe to SVN to Git.

C / C++ / Objective-C / C# / PHP / WinAPI / WTL / UIKit / SIP / Speex / AMR

EDUCATION, PATENT, LANGUAGES

Education	Engineer's degree in automation, 1999-2004, Ivano-Frankivsk National Technical University of Oil and Gas
Patent	Co-inventor of US patent 10,764,056 B2 "Short-distance network electronic authentication" (2020)
Languages	Ukrainian (native), English (B1-B2)